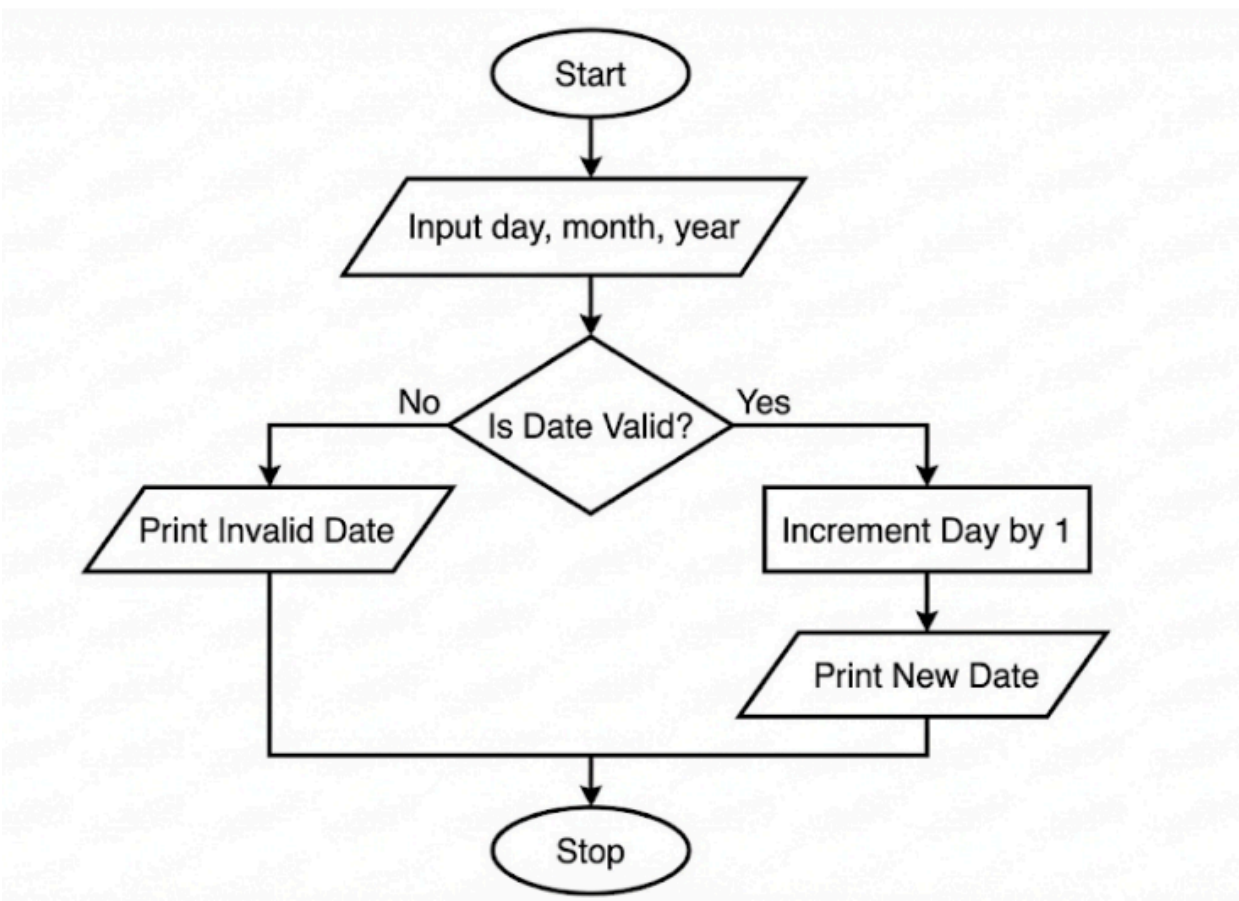
Experiment 6.1: Incremented Date

Algorithm:

```
Step 1: START
Step 2: INPUT day, month, year  (three separate integers)
Step 3: IF year <= 0 OR month < 1 OR month > 12
            → OUTPUT "Invalid Date" → STOP
Step 4: DETERMINE max_day:
           month in {1,3,5,7,8,10,12} → max_day = 31
           month in {4,6,9,11}        → max_day = 30
           month == 2:
               IF is_leap(year)        → max_day = 29
               ELSE                    → max_day = 28
Step 5: IF day < 1 OR day > max_day
            → OUTPUT "Invalid Date" → STOP
Step 6: INCREMENT:  day = day + 1
```

```
           IF day > max_day:  day = 1 ;  month = month + 1
               IF month > 12:  month = 1 ;  year = year + 1
Step 7: OUTPUT date as "DD-MM-YYYY"
Step 8: STOP
```

Flowchart:



Code:

CODETANTRA Home

pranjal.choudhary.batch2025@sitnagpur.siu.edu.in Support Logout

6.1.1. Incremented Date

01:06

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:

Sample Test Cases +

Explorer

nextDate.py

Submit

```python
day = int(input())
month = int(input())
year = int(input())

def is_leap(y):
    return (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0)

if year <= 0 or month < 1 or month > 12:
    print("Invalid Date")
else:
    if month in [1, 3, 5, 7, 8, 10, 12]:
        max_day = 31
    elif month in [4, 6, 9, 11]:
        max_day = 30
    else:
```

Debugger

Average time	Maximum time
0.048 s	0.064 s
47.70 ms	64.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 48 ms Debug

Expected output	Actual output
15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 2 43 ms

Terminal Test cases

‹ Prev Reset Submit Next ›